

Hsuan-Min (Eric) Chou

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EDUCATION

Pennsylvania State University

Master of Science in Computer Science Engineering

08/2023 - 05/2025

University Park, PA

- **GPA:** 3.7/4.0
- **Coursework:** Computer Vision, Data Structures, Algorithms, Operating System, Computer Architecture, Software Security, NeuroSymbolic Learning
- **Leadership:** Engineering Career Envoy

National Central University

Bachelor of Science in Computer Science, GPA: 3.92/4.3

09/2018 - 06/2022

Taoyuan, Taiwan

- **Coursework:** Data Structure, Algorithms, Operating System, System Programming, Programming Language Design, Data Science, Computer Organization, Computer Network
- **Awards:** Bronze Award - Ministry of Education AIoT Innovative Contest

WORK EXPERIENCE

SPARC

Software Engineer Intern

05/2024 - Present

State College, PA

- Developed a cross-platform Employee Relationship Management (ERM) application using **React**, **Node.js** and **Express**, enhancing employer-employee relations by **20%** in wellness and performance
- Designed **25+** mobile and web screens using **Figma**, improving UI/UX
- Built **20+ REST APIs** to support authentication with JWT and OAuth, integrating **MongoDB** to streamline workflows by **50%** in data storage and retrieval
- Deployed application infrastructure on **AWS EC2** and **S3**, utilizing PM2, achieving **99.9%** uptime

Intel

Software Firmware Engineer Intern

01/2022 - 07/2022

Taipei, Taiwan

- Developed an IP Simulation tool tailored for Linux system firmware development using **C** and **C++**, seamlessly integrated with **VS Code**, **CMake**, and **Google Test**
- Implemented over **15** functions and features into the tool, specially designed to streamline debugging process and support unit testing, resulting in a **10%** efficiency optimization
- Significantly improved code quality by increasing tool code coverage from **52%** to **95%** within one month

Barco NV

Software Engineer Intern

07/2021 - 10/2021

Taipei, Taiwan

- Built an automatic **NLP** NER-model with **Active Learning** algorithm to do text classification on **3700+** product reviews with sentiment analysis, within an Agile development environment
- Implemented manual correction with LabelStudio for data labeling to improve model accuracy at about **98.5%**, precision at 97.8%, recall at 98.6%, F1-score at 98.2%
- Developed a product reviews dashboard of 6 individual sentiment tags, utilizing **influxDB** and **Grafana** to track daily tagging reports, and **CI/CD** pipeline on Gitlab to automate daily training triggers

PROJECTS

Name-based Login IoT System

- Built a name-based login IoT system with **ESP32** and RFID sensor to scan users' UIDs, managing entry/exit status for over **400+** users
- Established a **Firebase NoSQL** database for managing user data, ensuring data retrieval and storage
- Developed a graphical interface for admin, improving data access and management efficiency by **30%**

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, C, C++, Java, HTML, CSS, SQL, $\text{\LaTeX}$

Web Development: React.js, Node.js, Express, MongoDB, Firebase, REST API, OAuth, JWT, Figma

Cloud & DevOps: AWS (EC2, S3), Git, GitLab CI/CD, Postman, Linux, Bash, Shell, CMake, Agile, Scrum